

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations**Supplementary Examinations - May - June 2023****20MEE73 - TOTAL QUALITY MANAGEMENT**

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of **SQC Table** is permitted.

Unit – I

Marks BT* CO* PO*

1. a) Explain the concept of TQM. List benefits of TQM. 10 L*1 1 1
- b) Explain the 7 basic tools used in quality control. 10 L2 2 1
2. a) Write brief notes on the following. i) Kaizen ii) Six Sigma 10 L2 2 1
- b) Define control chart. Explain the control chart for variables and attributes with example. 10 L2 2 1
3. a) Write brief notes on the following. i) ISO-9000 ii) ISO-14000 10 L2 1 1
- b) Determine mean, median mode and standard deviation for the given data set of a turning operation. Also find the number of components having more than 5mm diameter.

Diameter	2.7-2.9	3-3.2	5.1-5.3	3.6-3.8	4.5-4.7	4.8-5.0	3.3-3.5	3.9-4.1	4.2-4.4
Frequency	2	16	4	88	71	22	46	136	113

10 L3 2 2

Unit – II

4. a) Explain Type I and Type II error with example. 10 L1 3 1
- b) The table given below represents a listing of number of mechanical defects observed on lots of 10 miniature worm gears driven down daily from the line. This is a pilot production run lasting for 20 days. Construct a c chart and plot the points. Suppose the reason for out of control situation are known, what is your decision regarding the control limits for the future use?

Lot No.	No. of objects	Lot No.	No. of objects
1	13	11	17
2	15	12	11
3	19	13	7
4	8	14	11
5	6	15	14
6	17	16	6
7	7	17	16
8	9	18	10
9	3	19	2
10	23	20	6

10 L3 4 2

5. a) The following are the $\bar{X}$ and R values for 20 subgroups of 5 readings. The specifications for the product characteristics are 4.037 ± 0.0010 . The values given are the last two figures of the dimension reading; That is 31.6 should be 0.40316.

Subgroup	$\bar{X}$	R	Subgroup	$\bar{X}$	R
1	34.0	4	11	35.8	4
2	31.6	4	12	38.4	4
3	30.8	2	13	34.0	14
4	33.0	3	14	35.0	4
5	35.0	5	15	33.8	7
6	32.2	2	16	31.6	5
7	33.0	5	17	33.0	5
8	32.6	13	18	28.2	3
9	33.8	19	19	31.8	9
10	37.8	6	20	35.6	6

P.T.O.

- i) Determine the control limits for $\bar{X}$ and R charts for the future use, eliminating all the out of control points.
 ii) Will the process be able to meet the specifications?
 iii) What is R. P. I of the process?
 iv) Will you recommend shifting of the process centering?
- b) The following data refer to production and number of rejects for 15 consecutive days.

Day	Number inspected	Number of rejects
1	400	2
2	400	5
3	400	0
4	400	14
5	400	3
6	400	0
7	400	1
8	400	0
9	400	18
10	400	8
11	400	6
12	400	0
13	400	3
14	400	0
15	400	6
Total	6000	66

Plot 100p or percentage rejected control chart. Suppose the reasons for out of control points on 9th day is found and its effect on future production is eliminated what control limits will you adopt for future production.

6. a) Write short note on process capability index and process performance index & process capability ratio.
 b) An item is made in lots of 200 each and it has undergone 100% inspection. In the first 25 lots it was found that 75 items were found defective.
 i) Determine a trial control limit for up chart
 ii) Assuming all the points are within control limit, what is the estimate of process average fraction defective (p)?
 iii) If this p remains unchanged what is the probability of 26th lot will contain exactly 7 defectives? That it will contain 7 or more defectives?

Unit – III

7. a) What is the purpose of acceptance sampling?
 b) Write short note of operating characteristic curve.
 c) A double sampling plan is as follows.
 i) Select a sample of 2 from a lot of 20. If both are defective, reject the lot. If 1 is good and 1 is defective, take a second sample of one article.
 ii) If the article in the second sample is good accept the lot. If it is defective, reject the lot.
 If a lot is 25% defective is submitted what is the probability of acceptance? Compute this by method that is theoretically correct rather than by an approximate method.
8. a) The procedure of testing hypothesis requires a researcher to adopt several steps. Describe in all such steps.
 b) Define the following.
 i) AQL ii) LTPD iii) Consumer risk iv) Producer risk v) AOQL

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
